AKLT: separately proved entangled reference

H = sum of nearest-neighbor spin-2 projectors (coefficient 1)

Periodic gap ≥ 1/10 for every L ≥ 5

Unique ground state for L ≥ 3; complete omitted-state count = 3^L − 1

MPS bond D = 2 • One retained state • Local proof matrix 81 × 81

Finite-size projector criterion. Current thermal-transfer sufficiency fails.

This is not a thermal-bootstrap certificate or a bilinear-Heisenberg spectral-transfer result.